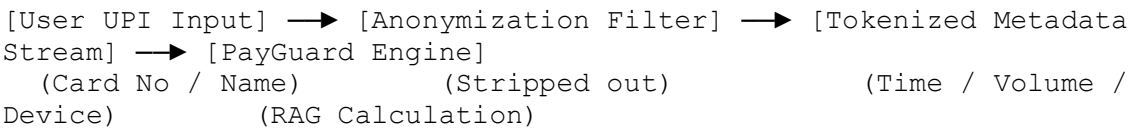


Secure Metadata Payment Flows & Tokenized Risk Architectures

1. Zero-Trust Metadata Architecture

Traditional payment fraud processing relies heavily on decrypting and viewing sensitive user inputs. PayGuard breaks this paradigm by employing a **Zero-PII Ingestion Pipeline**. This ensures the AI model operates strictly as a metadata validator.



1.1 Data Elimination Matrix

To protect customer privacy and maintain absolute compliance with international banking regulations, specific attributes are permanently scrubbed at the gateway layer before the transactional payload ever reaches the LLM context:

- **Primary Account Numbers (PAN / Card Details):** Completely stripped.
 - **Customer Names & Identifiers:** Anonymized into abstract alphanumeric session keys.
 - **One-Time Passwords (OTPs) / PINs:** Blocked globally at the interface layer.
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2. Telemetry Ingestion Points & Core Risk Scoring

The system parses exact structural indicators to identify potential account takeovers, fast wallet draining, and synthetic identity manipulation.

2.1 Core Threat Rules Reference Table

The agent evaluates the inbound metadata payload using three high-fidelity core rules. If an operational metric breaks a defined threshold, its corresponding risk factor is added to the session's cumulative score.

Rule Reference	Threat Classification	Target Metric	Risk Weight Value
Rule 101	Time Threat	Transaction executed between 00:00 and 04:00 AM	+30
Rule 102	Velocity Threat	Value exceeds 4x historical wallet daily average	+35
Rule 103	Identity Shift	Unregistered IMEI/Hardware ID or new IP location	+22

2.2 Mathematical Example Calculation

- Scenario:** A user requests a high-value transfer from a brand-new device at 2:00 AM.
 - Calculation:** Rule 101 (+30) + Rule 102 (+35) + Rule 103 (+22) = **87**
 - Result:** The transaction immediately crosses the critical high-threat boundary.
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3. Advanced Telemetry Extensions (Contextual Upgrades)

To catch sophisticated attack vectors such as SIM-swapping, automated script attacks, and remote access scams, the following evaluation layers are actively processed:

3.1 Rule 201: Geo-Velocity Anomalies

- **Trigger Condition:** The transaction originates from an IP address or cell-tower geolocation that is physically impossible to reach from the account's previous transaction location given the elapsed time window.
- **Metric Threshold:** Travel displacement vector exceeding **500 miles per hour** (e.g., a transaction in Mumbai followed by a transaction in London 45 minutes later).
- **Risk Multiplier:** **+40** risk score. Instantly overrides the transaction state to a minimum of Medium Risk.

3.2 Rule 202: Behavioral Biometrics & Session Telemetry

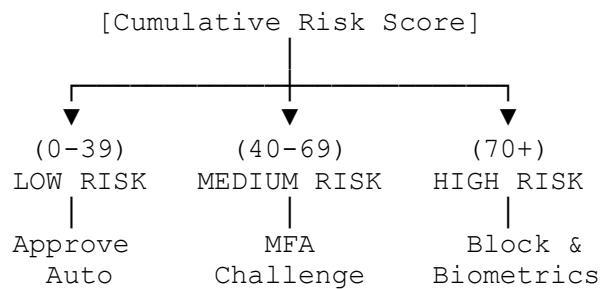
- **Trigger Condition:** Anomalies detected in how the user interacts with the application interface during payment setup.
- **Metric Threshold:** Typing cadence, keystroke flight time, or screen touch pressure deviates more than 60% from the historical user profile, or copy-paste actions are used to enter payment amounts instead of manual numerical typing (indicates automated script injection).
- **Risk Multiplier:** **+18** risk score.

3.3 Rule 203: High-Risk Destination Routing

- **Trigger Condition:** The recipient endpoint is matched against a known high-volatility financial node or immediate settlement network.
 - **Metric Threshold:** Funds are routed directly into newly generated peer-to-peer (P2P) wallets, instant-mixing crypto currency exchanges, or unverified commercial payment gateways with a domain age of less than 30 days.
 - **Risk Multiplier:** **+25** risk score.
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4. Mitigation Matrix & Enforcement Flows

Cumulative scores are continuously calculated and matched directly to a strict corporate risk containment playbook. This ensures instant enforcement across all connected banking nodes.



4.1 Low Risk Profile (Score Range: 0 - 39)

- **Operational Directive:** Clear for immediate clearing.
- **Action:** The system bypasses active hurdles, updates the local historical wallet average, and transparently routes the payload to the ledger network.

4.2 Medium Risk Profile (Score Range: 40 - 69)

- **Operational Directive:** Intercept and challenge.
- **Action:** Settlement is placed on a temporary hold. The core engine fires an automated Multi-Factor Authentication (MFA) webhook back to the user's primary registered device, requiring push-notification confirmation.

4.3 High Risk Profile (Score Range: 70+)

- **Operational Directive:** Defend and freeze.
 - **Action:** The payment flow is immediately aborted. The platform drops the routing session, locks down wallet velocity limits for 24 hours, and completely restricts fallback text-message OTP verification (to prevent bypassing via SIM-swap cloning). The account requires out-of-band biometric authentication (facial recognition or fingerprint telemetry) to unlock.
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5. System Interoperability & API Error Controls

When integrating with external core banking systems via webhooks, the following telemetry status codes must be returned to ensure predictable AI agent parsing:

- **HTTP 200 (Success - Code PG-01):** Evaluation complete. Risk indicators parsed successfully and mitigation score mapped.
 - **HTTP 403 (Blocked - Code PG-99):** High-risk critical threshold breached. Settlement frozen dynamically.
 - **HTTP 422 (Unprocessable - Code PG-ERR):** Input contains structural PII violations (e.g., raw credit card text passed). The pipeline triggers an instant security fallback, scrubs the logs, and forces a hard reset of the agent's interaction session.
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